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The Editor-in-Chief
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Dear Editor,

I am submitting the enclosed research paper, “Cross-species separability of radar cardiorespiratory signatures is no larger than cross-hardware separability: a leakage-aware re-evaluation of three public datasets”, for consideration as a Research Paper in Physiological Measurement.

The paper asks a measurement question the radar vital-sign literature has not settled: once window overlap is removed and the sampling grid is matched, how much cardiorespiratory “biometric” information remains, and is it larger than what the same pipeline extracts from a change of hardware within one species? Three public datasets (30 dogs at 7.3 GHz, 110 adults at 60 GHz, 24 adults at 120 GHz) pass through one 20 Hz chain. The answer is negative and reproducible: closed-set identification of 30 dogs falls from 54.4 % under random-window cross-validation to 46.1 % under a subject-grouped temporal split and 16.3 % at a 25 min horizon; open-set verification falls to chance (AUROC 0.52–0.59) across 20 random partitions; and dog-versus-human balanced accuracy (86.2 %) is smaller than the 93.8 % obtained for the same adults recorded on two different radars.

I believe this fits the journal's emphasis on quantitative assessment of physiological function and on the validation of measurement methods. The contribution is methodological and cheap to adopt: a ladder of three validation protocols applied unchanged to identification, open-set verification and cross-species classification; three same-data negative controls that bound what the pipeline can extract without any species difference; a rate-matched decimation control that separates descriptors encoding the sampling grid from those that do not; and a transfer test in both directions between platforms. None requires new hardware or recordings.

The negative result is the point, not a by-product. High cross-species and open-set accuracies are widely quoted in this field, and the controls reported here show that such numbers can be explained by acquisition condition alone. Reporting that honestly requires an inspectable analysis, so the complete pipeline, the feature tables and every result file are public at https://github.com/Xbotsd/Radar_Cardiorespiratory_Biometrics under the MIT licence, with 39 tests and a generated table mapping each quoted number to its result field.

Two disclosures. First, the 120 GHz dataset contributes 24 participants, below the journal's guideline of 30; it is a secondary analysis of a published dataset and contributes no primary estimate, appearing only in the vital-sign agreement table, the hardware negative control and the transfer test, while every primary human result uses the 110-adult 60 GHz cohort. Second, an earlier analysis of the same datasets was prepared for two IEEE journals and was not accepted; the present manuscript is a new study, adding the leakage ladder, the three negative controls, the rate-matched ablation, the bidirectional transfer test, the age negative result and the public, tested code base. Neither earlier manuscript is under consideration anywhere and no part of this work has been published.

The manuscript is not under consideration by any other journal. All data are public and de-identified; no new human or animal subjects were involved, and the original publications report the consent and animal-care procedures under which the recordings were made. I declare no competing interests and no specific funding.

Yours sincerely,

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